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NITTE (Deemed to be University)

(Established under Sec.(3) of UGC Act, 1956)

First Semester (CBCS) Degree Examinations, NOV. - DEC. 2025**25CHY104– MATERIALS CHEMISTRY FOR DEVICES AND E-WASTE MANAGEMENT**

(FOR ACT, AID, AIM, CBS, CCE, CSE, ECE, EEE, ISE, RAI, VDT)

Duration: 3 Hours]

[Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **One full** question from each Unit.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. A working concentration cell is made of two half cells having
A) Identical electrolytes and identical concentrations
B) Identical electrolytes but different concentrations
C) Different electrolytes and identical concentrations
D) Different electrolytes and different concentrations
2. is used as anode material in commercial lithium-ion batteries.
A) Lithium intercalated in Graphite layers
B) Silicon
C) Lithiated metal oxides
D) Platinum
3. A glass electrode is commonly used to measure
A) Oxidation potential
B) Reduction potential
C) pH of a solution
D) Conductivity of a solution
4. The EMF of the concentration cell $\text{Ag(s)}|\text{Ag}^+(0.05\text{M})||\text{Ag}^+(0.05\text{M})|\text{Ag}$ at 298°K is
A) 0.0591V
B) 0.02955V
C) 0V
D) -0.0591V
5. A steel tank fitted with a copper tap undergoes corrosion in water because
A) Electrode potential of both metals is almost equal
B) Steel acts as an anode and corrodes due to galvanic action
C) Steel is susceptible to water
D) Copper acts as an anode and corrodes
6. Which of the following is not a corrosion prevention technique?
A) Sacrificial Anode
B) Protective coating
C) Pickling
D) Anodizing
7. In the electroplating process, generally the object to be plated is made as
A) Anode
B) Electrolyte
C) Cathode
D) Power house
8. In the production of double-sided printed circuit boards, connection-through-hole is efficiently made by
A) Electroplating
B) Electroless plating
C) Molten dip method
D) using copper wires
9. During conductometric titration of a mixture of strong and weak acids against NaOH, which of the following is true?
A) Weak acid will be neutralized first
B) Strong acid will be neutralized first
C) It depends on the relative volumes of the strong and weak acids taken
D) It depends on the relative concentrations of the strong and weak acids taken
10. A PNP junction is normally found in an electronic memory device.
A) Transistor-type
B) Capacitor-type
C) Resistor-type
D) Charge-transfer-type

11. A numeric liquid crystal display has segments.
 A) Thirty-two B) Sixteen
 C) Eight D) Seven
12. Working principle of LED is
 A) Photoluminescence B) Chemi-luminescence
 C) Electrooptic effect D) Electro-luminescence
13. Polymers that soften on heating and harden on cooling are called
 A) Thermoplastics B) Elastomers
 C) Thermosetting polymers D) Resins
14. Why does undoped polyacetylene behave as an insulator?
 A) It has a high density of free electrons B) It is not crystalline
 C) It has a relatively large bandgap D) It lacks extensive conjugation
15. Photolithography technique for synthesis of nanomaterials makes use of
 A) UV radiation B) IR radiation
 C) Visible radiation D) Solar radiation
16. Which of the following property is responsible for making nanomaterials good catalysts?
 A) Increased volume B) Increase in melting point
 C) Increase in surface area to volume ratio D) Increased conductance
17. Hydrometallurgy involves treating e-waste with
 A) Water B) Leaching agents
 C) Organic solvents D) Ionic liquids
18. Magnetic separation of pretreatment enables separation of
 A) Noble metals from plastics B) Glass wastes from plastics
 C) Non-ferrous metals from plastics D) Ferrous metals from non-ferrous metals
19. Which leaching agent is commonly employed for gold recovery from e-waste?
 A) Aqua regia or cyanides B) Strong ammoniacal solution
 C) Mild organic acids D) Mixture of ionic salt and weak base
20. Cadmium present in e-waste mainly affects
 A) Kidney and nervous system B) Bone and tooth
 C) Skin D) Digestive system

PART - B: DESCRIPTIVE ANSWER QUESTIONS

80 Marks

Unit – I		Marks	BT*	CO*	PO*
1.	a) Derive an equation for glass electrode potential how glass electrode is used in the determination of pH.	8	2	1	1
	b) A concentration cell was constructed by immersing two copper electrodes in 0.02 M and 2 M CuSO_4 solutions. Write the cell representation, cell reactions, and calculate the EMF of the cell at 25 °C.	4	3	1	2
	c) Differentiate primary and secondary batteries.	4	2	1	1
2.	a) Explain the construction, working, and two applications of a lithium-ion battery.	7	2	1	1
	b) Discuss the construction and working of a photovoltaic cell.	5	2	1	1
	c) Calculate the cell potential and write the cell reactions of the following cell at 298K $\text{Ni} \text{Ni}^{2+}(0.001\text{M}) \text{Ni}^{2+}(0.50\text{M}) \text{Ni}$.	4	3	1	2
Unit – II					
3.	a) Explain the electrochemical theory of corrosion with reference to iron.	7	2	2	1
	b) A 5.5 kg iron metal piece of density 8.1g/cm^3 with an area of 10.0 inch^2 was submerged in seawater for 10.5 years. The weight of the remaining metal was found to be 2.4 kg. Calculate CPR in mpy and mmpy.	5	3	2	2

c)	Mention the two advantages and two disadvantages of electroless plating.	4	1	2	1
4. a)	Explain the electroplating process of decorative and hard chromium.	7	2	2	1
b)	A piece of bronze with a surface area of 50 inch ² was exposed to a corrosive environment for a period of 2 years. It had experienced a weight loss of 200g. Calculate CPR in mpy and mmpy. Given: density of bronze is 8.8 g/cm ³	5	3	2	2
c)	Define galvanization. Explain how it helps in preventing corrosion of iron.	4	1	2	1

Unit – III

5. a)	Describe principle and procedure for the potentiometric technique for estimation of Ferrous Ammonium Sulphate (FAS) using standard K ₂ Cr ₂ O ₇ .	7	2	3	1
b)	Explain the construction and working of a Liquid Crystal Display (LCD).	5	2	3	1
c)	Define transistor and capacitor-type memory devices.	4	1	3	1
6. a)	Explain the construction, working, and applications of organic Light Emitting Diodes (OLEDs).	7	2	3	1
b)	Analyze the application of an electrochemical gas sensor in the detection of sulfur dioxide (SO ₂), with relevant reactions.	5	2	3	1
c)	List the 4 advantages of organic molecules in electronic memory devices.	4	1	3	1

Unit – IV

7. a)	Explain the conduction mechanism in polyacetylene when it is doped with n-type and p-type dopants.	7	2	4	1
b)	A sample of polyethylene consists of 200 molecules of molecular mass 500 g/mol, 300 molecules of molecular mass 2000 g/mol, and 500 molecules of molecular mass 3500 g/mol. Calculate the number average molecular weight and weight average molecular weights of a polymer.	5	3	4	2
c)	"Catalytic property of nanomaterials differ from their bulk materials" Justify your answer using examples.	4	2	4	1
8. a)	Explain photolithography technique for synthesis of nanomaterial	7	2	4	1
b)	A sample of polystyrene consists of: 150 molecules of molecular mass 600 g/mol, 400 molecules of molecular mass 1800 g/mol and 450 molecules of molecular mass 3600 g/mol. Calculate the number-average molecular weight, and weight-average molecular weight of this polystyrene sample.	5	3	4	2
c)	Differentiate between thermo-polymers and thermosetting polymers	4	2	4	1

Unit – V

9. a)	Explain the stages involved in the process of gold recovery from e-waste.	7	2	5	1
b)	Illustrate how ion exchange method helps in extracting metals from e-waste.	5	3	5	1
c)	List the steps involved in the pretreatment of e-waste before metal extraction	4	1	5	1
10. a)	Define pyrometallurgy. Explain any three pyrometallurgical methods used for the extraction of metals from e-waste.	7	2	5	1
b)	Explain acid leaching and cyanide leaching used in extraction of metals.	5	3	5	1
c)	Mention the sources and composition of the e-waste.	4	1	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
